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Homework 3
RBIF 101

Question 1:

For the homology modeling, I ended up using Phyre2, RosettaCM (via Robetta), and HHPred + MODELLER (via the MPI Bioinformatics Toolkit) [1-5]. To get an hGPR91 sequence, I went to the NCBI website and searched the Protein database for "GPR91", which led me to the reference sequence for human succinate receptor 1: NP_149039.2 (aka hGPR91/hSUCNR1) [6]. Due to how Phyre2 works, a different template had to be selected for the homology modeling using that tool compared to the other two. Both templates used were rat SUCNR1 structures with sequences that were ~95% identical, so these templates were quite comparable. The rationale for selection of either template is discussed in detail below

1(a):

Template selection:

Phyre2 performs its homology detection to generate a list of templates, from which it automatically chooses one. Upon closer inspection, the chosen Phyre2 template did have the best combination of confidence score, alignment coverage, and percent identity with the target sequence from the list of options. As such, I decided to use the model built using the automatically selected template. More specifically, the template chosen by Phyre2 was chain C of 6IBB, a structure of GPR91 from rat [7]. This seemed like a decent template given that it is a relatively close ortholog of hGPR91 and had 56% percent identity with the target over ~284 aligned residues (85% coverage, according to Phyre2). This combination of percent identity and alignment length fell within the "safe" range for homology modeling according to the chapter on this topic by *Krieger et al.* from *Structural Bioinformatics* (see Figure 25.1) [8]. The rest of the templates detected by Phyre2 were borderline at best according to the same guidelines from Chapter 25 of *Structural Bioinformatics* [8]. I will say that the resolution of 2.12 Å and overall quality assessment for 6IBB on the PDB were not ideal, or as good as that of the other template discussed below. That said, 6IBB-C was the best available option offered by Phyre2.

I used Robetta in comparative modeling mode (i.e. RosettaCM) [3], which allows you to provide a PDB file as a template structure. To identify a suitable template, I used BLASTP to search for the target's sequence (NP_149039.2) against the sequences in the PDB [9]. I ignored several top hits that were human or synthetic construct sequences with 100% identity and coverage with the target. That said, there was a good homology hit from a rat SUCNR1 structure: chain A of 6RNK [10]. This BLASTP hit had 94% coverage, an E-value of 1e-176, and 75.24% identity with the query. Even factoring in a drop in percent identity for a global alignment of the target with this same template, the >300 residue local alignment length with 75% identity seemed to easily put this template in the safe zone for accurate homology modeling offered by *Krieger et al.* [8]. The idea is that because structure is conserved over much longer timescales than sequence, for sequences that are sufficiently identical or similar the probability of strong structural similarity becomes quite high, making such homologs the safest choices as templates [8]. I fetched 6RNK in PyMol and exported a PDB file with just chain A for use with Robetta. Upon upload of that structure, Robetta calculated an initial global alignment with a length of 334 and a percent identity of ~66% (still acceptable) [8]. Also, while it would ideally be better, the quality of this template structure was passable, with a 1.94 Å resolution and better-than-average assessment for most PDB validation metrics.

The MPI Bioinformatics Toolkit puts many useful tools together in one web service [4]. Through that website, I used HHPred to conduct an HMM profile-based homology search for my hGPR91 target (NP_149039.2) against a subset of the PDB [4]. This led to a generous list of potential templates, in which 6RNK-A from rat was again the

best non-identical/non-human hit. Because I already knew this hit to be a promising template based on the considerations discussed above for RosettaCM, I proceeded with this structure, which was sent to MODELLER after the alignment of the template and target via the MPI Bioinformatics Toolkit.

Methodology differences:

Phyre2 was developed with the detection of more distant homologs in mind, for scenarios where templates with high percent identities with the target are not necessarily available [2]. While the Phyre2 server can be used in a variety of ways, the standard pipeline that I used involves first running the target sequence through HHBlits [2]. HHBlits is a remote homology detection software that, much like other tools of this type, uses pairwise alignments of HMM profiles for groups of homologous proteins, with some heuristics and algorithmic optimization to dramatically speed up the search [2,11]. PSIPRED, a neural network trained on sequence profiles, is also used to predict the location of secondary structural elements within the target based on the profile generated by HHBlits [2]. The predicted secondary structural elements and HHBlits profile for the target are combined into another HMM, which is compared to HMMs in a "fold database" using HHsearch, finally yielding a list of hits ranked by posterior probabilities [2]. The hits from this library are used to create rough backbone models for the target that have insertions and deletions [2]. The insertions and deletions are resolved by finding the most plausible fragments from a library of fragments from real proteins and putting them into the crude models via a dihedral angle optimization procedure. These models are ranked by their plausibility using empirical energy-based comparisons and the top model is chosen [2]. Side chains are added to the chosen model from a rotamer library using a graph-based algorithm to place them in the most likely orientation with no clashes [2].

I used the **RosettaCM** (comparative modeling) pipeline when using Robetta. The user provides templates to RosettaCM, and I found a template from a rat ortholog via BLASTP (described earlier). PSI-BLAST was not necessary in this case because suitable templates could be detected easily. RosettaCM first involves threading the target onto the template(s) provided to generate crude and incomplete models, which are then all structurally aligned [3]. These models are then used as the scaffold (via distance constraints) for a Monte Carlo sampling procedure guided by an energy function and spatial gap penalty [3]. This sampling procedure uses fragments from the template(s) and ones generated by Rosetta from PDB data based on the local sequence of the target [3]. This typically leads to the correct general topology, with further refinement conducted by Monte Carlo sampling to find more optimal fragments and conformations, prioritizing regions with poor local geometry [3]. Approximated energy optimization is conducted to adjust the entire refined structure, and then side chains are added and optimized via a more realistic energy function [3].

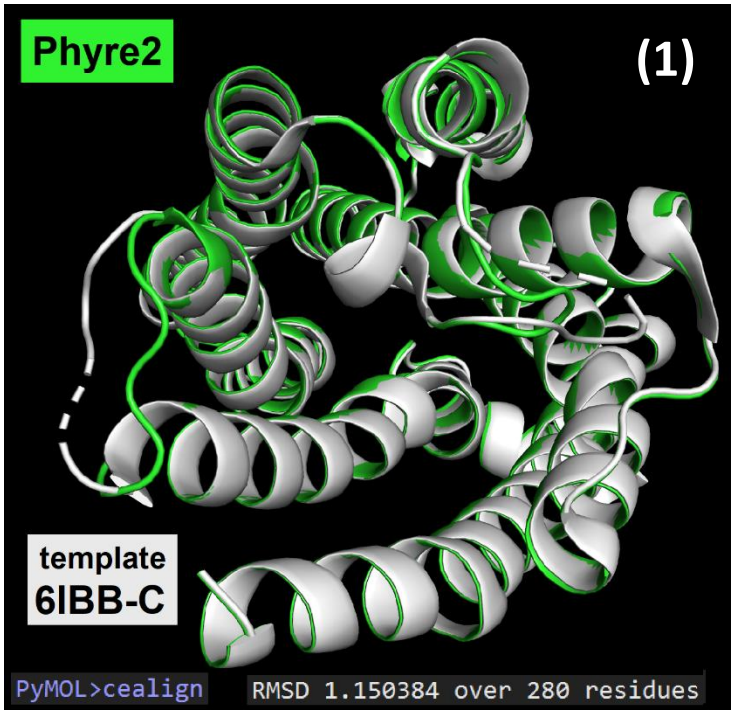
When using the MPI Bioinformatics Toolkit, **HHPred** was only used for the homology search (i.e. for template selection), while **MODELLER** was used for the actual prediction of target structure using the template [4]. HHPred first detects homologs as potential templates by HMM-HMM profile comparison, and like Phyre2, it also uses HHBlits for this step [4]. The sequence alignment produced by HHPred of the target and the homology hits selected as template(s) can then be forwarded to MODELLER [4]. MODELLER uses the mapping of the target to the template to predict a structure by imposing a variety of constraints in the form of probability density functions estimated empirically from pairwise alignments of known structures [12]. These constraints take spatial considerations like bond lengths and dihedral angles throughout the structure into account, and MODELLER aims to find the solution with maximal probability given all constraints [12]. MODELLER can also be used to do other specialized tasks like predict loops *ab initio*, but I think I only used its core comparative modeling functionality via the MPI Bioinformatics Toolkit [4,12,13].

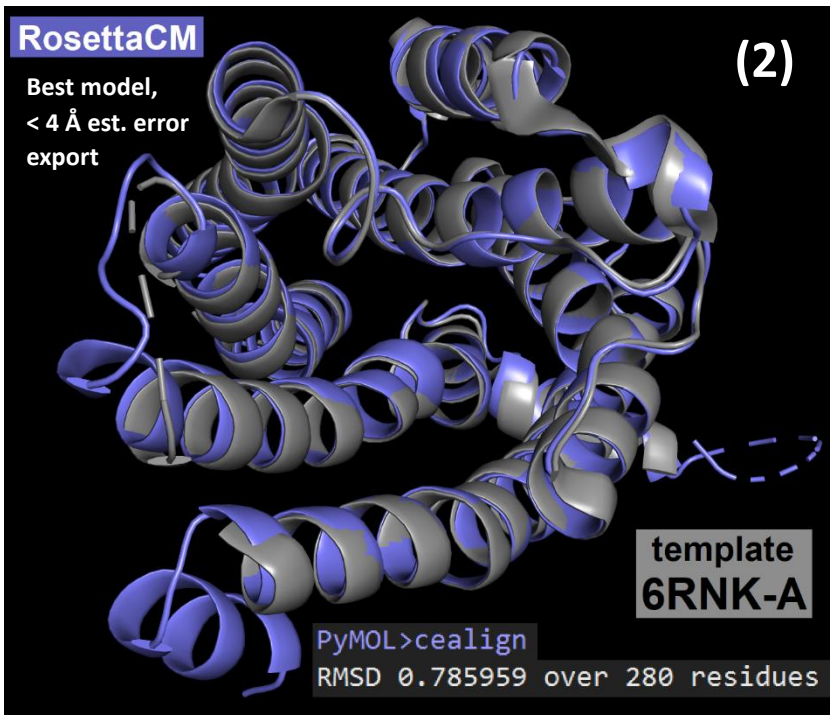
As I used them, all 3 of the homology modeling tools described in detail above **aim to achieve the same goal in slightly different ways**. At its core, Phyre2 seems to take a fold-recognition/threading approach, first identifying

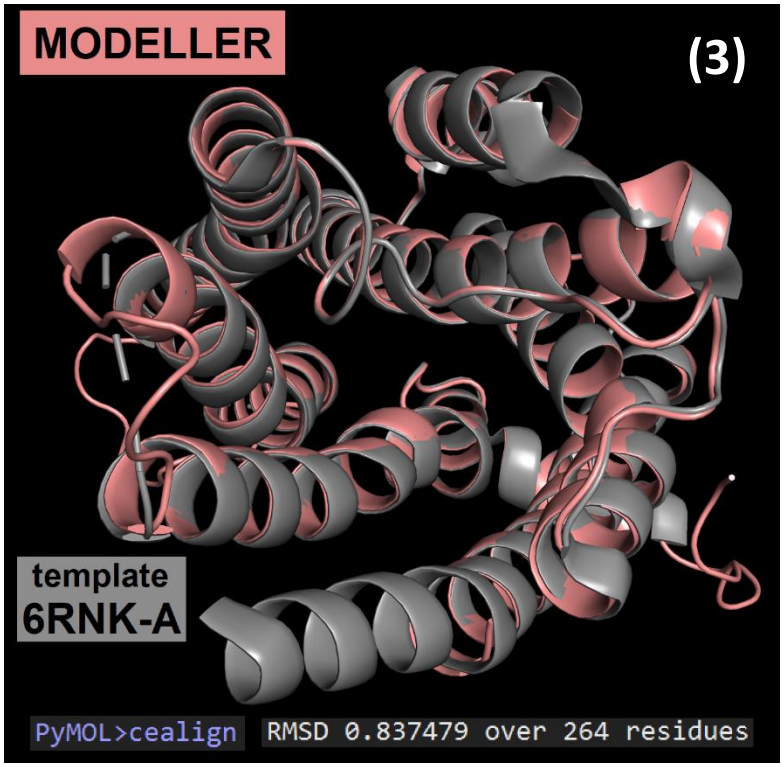
template structures from a library and then using them to create crude models for the target [2]. That said, Phyre2 does not stop after threading, and the rough models are improved by finding and using fragments from a library to resolve indels in the initial target models [2]. RosettaCM also appears to use threading to hammer out crude target structure(s) based on the provided template(s), which are then improved by methodically incorporating fragments [2,3]. For these reasons, at a high level Phyre2 and RosettaCM seem quite similar. That said, a notable difference comes in how the fragments are found and used [2,3]. RosettaCM's Monte Carlo fragment sampling procedures seem much more complicated, but also more flexible and robust, with fragments coming from the PDB and also generated *de novo* by Rosetta itself [3]. Unlike Phyre2, RosettaCM involves two rounds of improvement using fragments: a first pass to improve the threaded models and a second pass to fix geometrically problematic areas, some of which are left over by the first pass [3]. MODELLER is perhaps the most different from Phyre2 and RosettaCM, as it does not appear to use fold-recognition/threading or fragments to build the target model [5,12,13]. Instead of threading and improvement by fragment selection and placement, MODELLER uses the alignment with the template(s) as a spatial framework around which the target structure is built by optimizing within a set of spatial constraints, which are based on what we have observed with regards to bond lengths, atom distances, dihedral angles (backbone or side chain), etc. within proteins [12,13]. MODELLER essentially builds the target from the template by considering the statistics of the possible conformations of atoms within the protein [14]. In this regard, it seems that MODELLER uses a process that is a little bit closer to (although not the same as) explicitly modeling protein folding as a physicochemical phenomenon, particularly when compared to the other two methods.

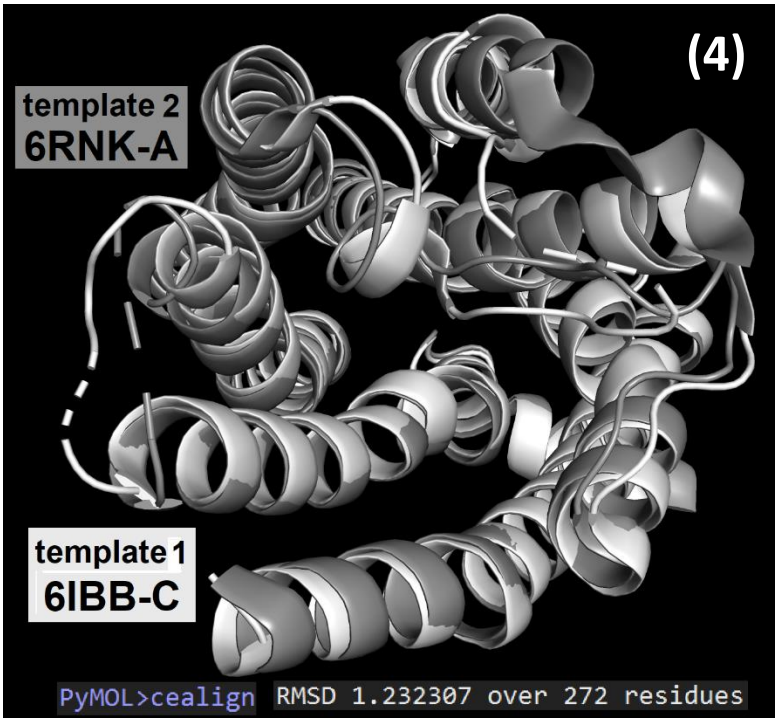
1(b):

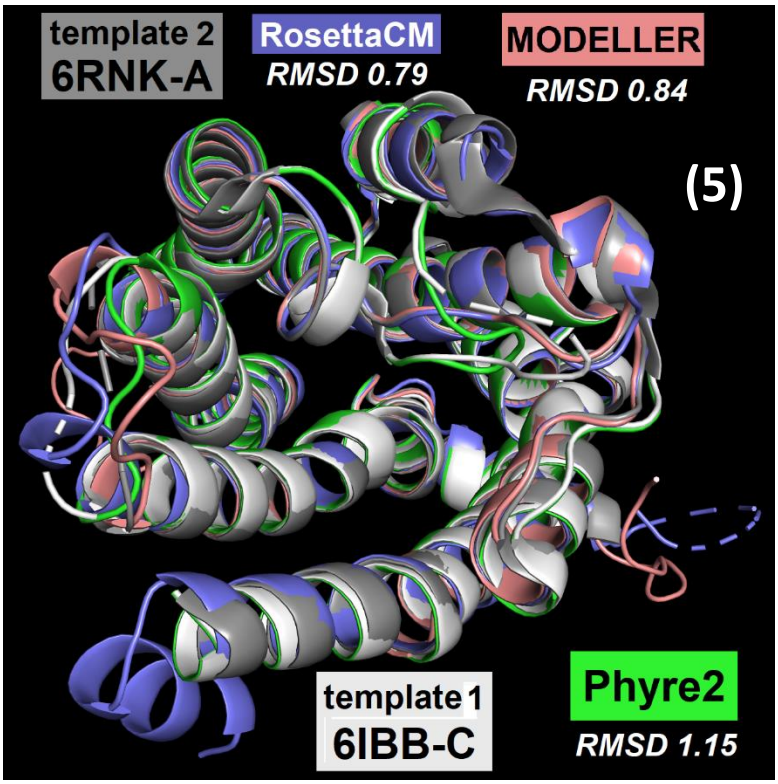
I had to use a different but very sequence-similar template for Phyre2 (6IBB-C) compared to the other two methods (6RNK-A). For this reason, images are included below and on the next page for each predicted model aligned to its respective template, as well as an image of all models together. In the image of all models together, models were aligned to their respective templates after the two templates were aligned to each other. For reference, on the next page there is also an image of the two templates alone and aligned to each other.









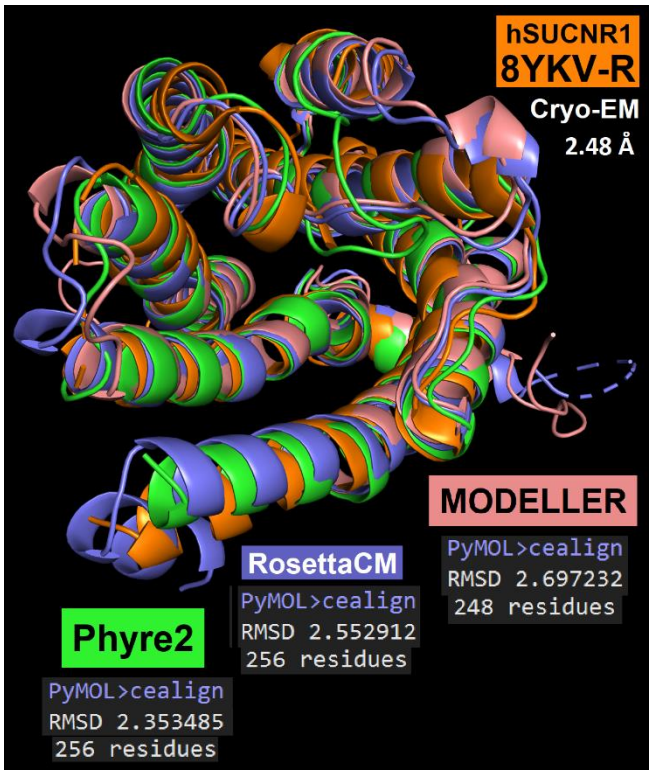


The hGPR91 models generated by **RosettaCM** (indigo) and **MODELLER** (salmon) aligned to their common template 6RNK-A (grey) with **RMSDs of 0.79 and 0.84, respectively (Figures 2, 3, & 5)**. The model generated by **Phyre2** (green) aligned to its template 6IBB-C (white) with an **RMSD of 1.15 (Figures 1 & 5)**. All alignment images look down onto the extracellular side of SUCNR1 with the succinate binding site, which is approximately situated behind and below the left side of the loop just above the images' center.

1(c):

It is worth noting that while both templates were from rat GPR91 and had around 95% sequence identity, these template structures are not quite identical, with an RMSD of 1.23 when aligned to each other (**Figure 4** above) [7,10]. Thus, some differences between the model from Phyre2 and the others may simply be attributed to differences in the templates used. Overall, the models generated by all three methods had excellent alignment to their respective templates around the core structure, i.e. the 7 transmembrane α-helices (qualitatively inspected in PyMol). Three of the shorter loops were even modeled quite similarly between the three methods (only one of these is visible above). All of this is consistent with the results in the study by *Costanzi*, in that template-based modeling generally does an excellent job at predicting the core fold/structure of a protein [15], i.e. how the positioning and configuration of major secondary structural elements come together. It is the loops, particularly longer ones, that have the most variability among homologs and are thus most challenging to model [15]. Qualitatively speaking, the result from MODELLER appears to have adherence to the template around the 7 transmembrane α-helices comparable to RosettaCM's model, meaning that RosettaCM's slightly better RMSD may be mostly due to increased overall adherence to the template's loops. MODELLER did

predict one loop quite differently from the template or RosettaCM, which is hidden from view on the opposite side of the structural alignments on the previous page. Another deviation in a disordered region can be seen on the right side of the image of all structures (salmon versus indigo). It is worth noting that MODELLER's model aligned with the template over fewer residues than the other two methods, and a section of missing α-helix (TM1) can be identified by comparing the MODELLER-only screenshot to the others (bottom of **Figures 1-3**). Phyre2 had the highest RMSD of the predicted models. It was noted earlier that the template structure used by Phyre 2 was of slightly lower overall resolution and quality than the one used by the other two methods [7,10], and this may have been a contributing factor. It is difficult for me to choose the "best" model with much confidence from among the 3 shown earlier. To me, the study by *Costanzi* highlighted that adherence to the template does not perfectly indicate accuracy relative to the true structure of the target (once solved) [15]. That said, I would tend to trust the results from RosettaCM and MODELLER more than Phyre2, not solely due to their lower RMSDs with the template, but also the better quality of the template structure itself according to the PDB validation reports [7,10]. Supporting this, template 6IBB got a MolProbity score within the 81st percentile of structures with comparable resolution, while 6RNK got a MolProbity score within the 96th percentile of comparable structures [16]. Regardless of the approach, the Week 5 reading materials taught me that homology modeling results can only be as good as the alignments and/or templates used. Without access to additional information, I would weigh these factors most strongly to guide my trust in a model with the understanding that some inaccuracies are unavoidable. As for RosettaCM versus MODELLER, I might lean toward RosettaCM due to the lower RMSD and higher alignment coverage with the template, as well as the reputation of Rosetta-based utilities with regard to performance (though MODELLER is also well-established).

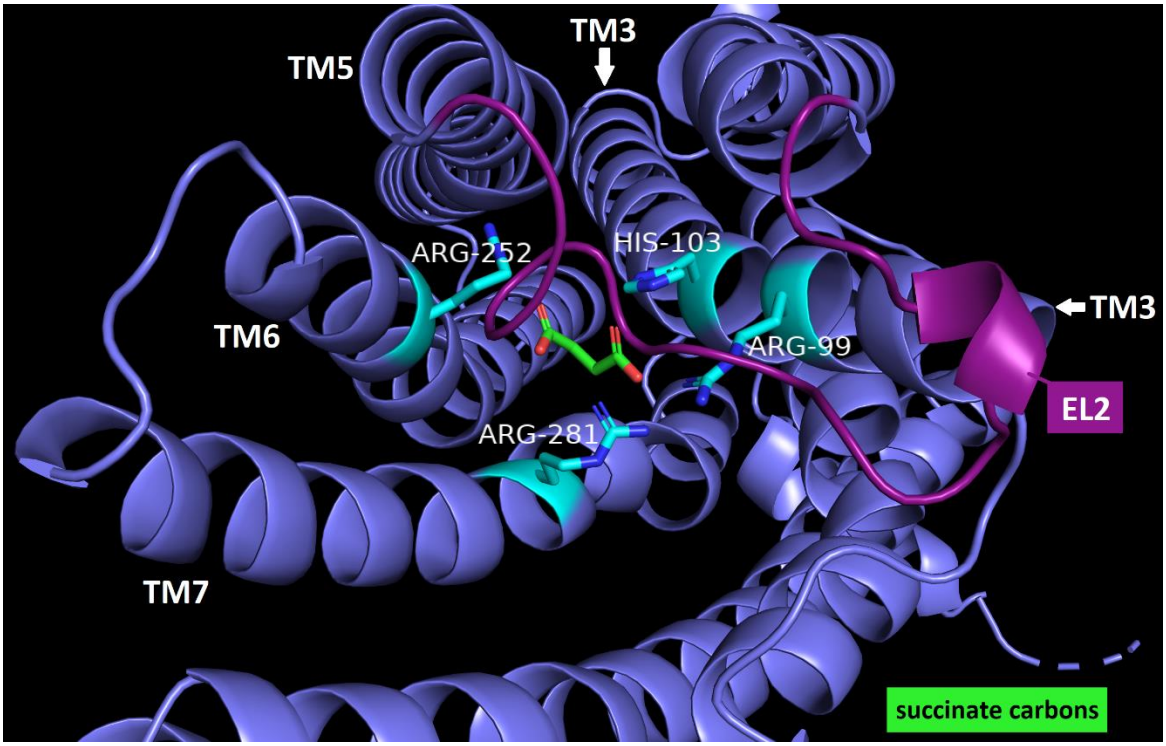


Given that some structures for hSUCNR1 now exist, I was curious to know which model was closest to an actual structure of the target. I fetched a structure with a sequence identical to my target sequence (8YKV-R) in PyMol and aligned the 3 homology models to it to see what the RMSDs were (cealign). While this may be cheating a bit because homology modeling is performed in situations where such information is not available, I was curious to check it out. This structure was identified during my earlier BLASTP homology search against the PDB, which led me to 6RNK-A. While the 2.48 Å resolution of this structure was not great (it's from Cryo-EM), the PDB quality report puts this structure among some of the best in the PDB [17]. The model from Phyre2 aligned to 8YKV-R with an RMSD of 2.35 over 256 residues, the one from RosettaCM had an RMSD of 2.55 over 256 residues, and the one from MODELLER had an RMSD of 2.7 over 248 residues (image to the right). Because of what I had discussed above, I was initially surprised that Phyre2 had the lowest RMSD with an actual structure of the target. However, given the resolution and uncertainty likely involved, the best that can probably be said from this inquiry with any confidence is that all 3 methods may have performed comparably well. It is also worth noting that 8YKV-R captures an ***activated*** *state of hSUCNR1, while the rat templates upon which these models were based capture an* ***inactive*** *state (6RNK-A and 6IBB-C) [18,19]. This could contribute to the higher RMSDs seen in the alignments to the hSUCNR1 structure above compared to the alignments to the templates shown earlier.*

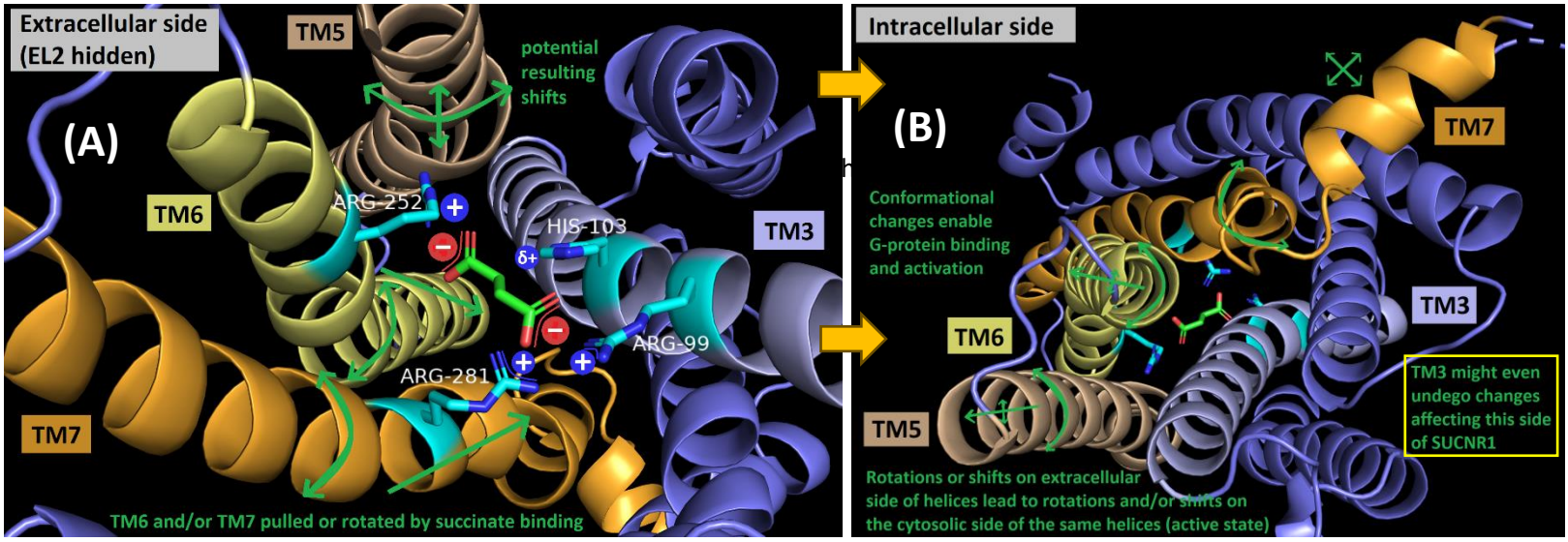
Question 2:

First, from a chemistry perspective, succinate can associate non-covalently with these residues, mostly notably via charge-based interactions. Due to the pKa of its two carboxyl groups (likely < 5), at physiological pH succinic acid will mostly exist as succinate [20-22]. Succinate has a net charge of -2 due to the loss of a hydrogen ion/proton from each of the hydroxyl groups, and each negative charge will be delocalized/shared between both carboxyl oxygens due to resonance [21,22]. So, the ligand has one full negative charge located at either end. Meanwhile, ARG-99, ARG281, and ARG-252 will each have a full positive charge at physiological pH [23]. This charge will exist delocalized across the two guanidino nitrogen atoms at the end of the side chain due to resonance [24,25]. HIS-103 will only be partially positively charged at physiological pH, and this charge will be delocalized across the nitrogen atoms in its imidazole ring via resonance [23,25]. Thus, the side chains of the arginine residues can interact with either end of succinate via electrostatics. The same is true for histidine, only the interaction is likely a bit weaker.

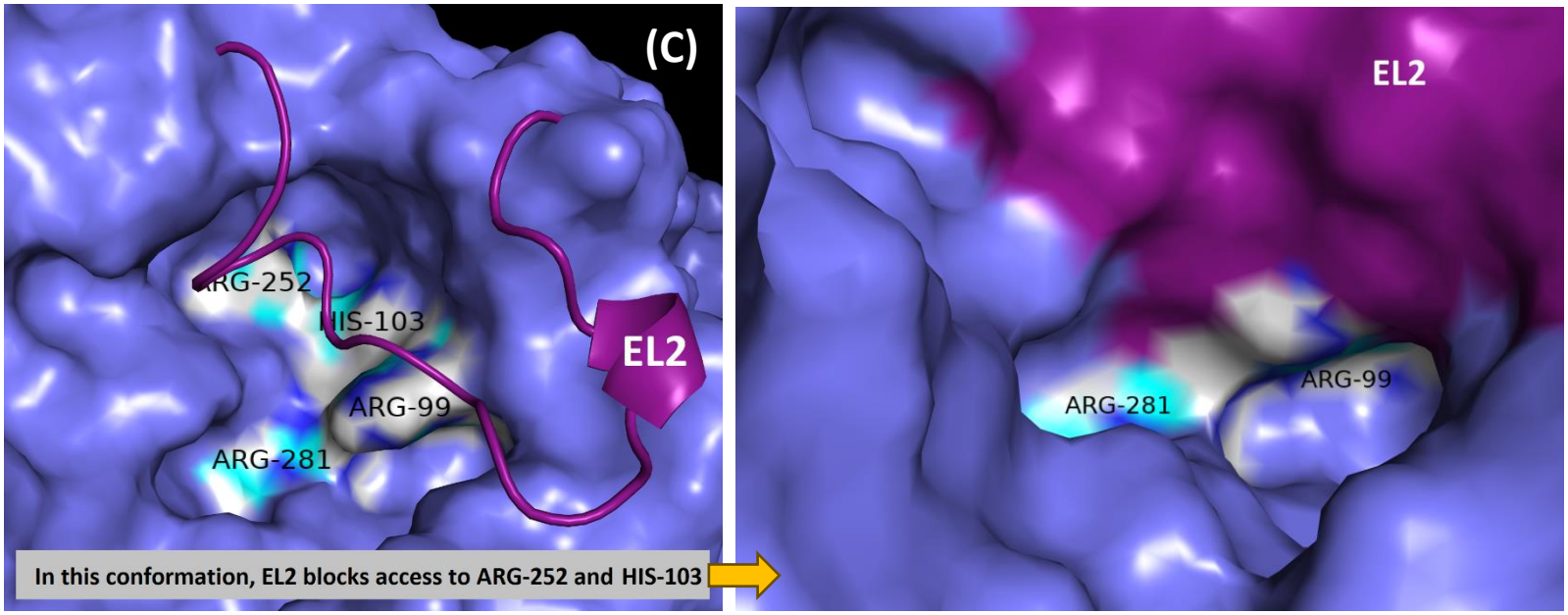
So, it is clear that succinate can interact with these residues via relatively strong electrostatics, but to what end? How might these residues contribute to the signal being "passed on" to the intracellular side of the hSUCNR1 receptor? Looking at a homology model generated in Question 1 can provide some hints as to what might be going on. For simplicity, I will only consider the model generated by RosettaCM from the rat SUCNR1 template (6RNK-A). The Phyre2 model had one of the arginine side chains placed and oriented unusually far away compared to the other models and the image in the HW3 instructions. The MODELLER output was comparable to that of RosettaCM around these residues, with subtly different side chain rotamers. Below is an image of the RosettaCM model predicted for hSUCNR1 in cartoon with the residues of interest labeled and their carbons colored cyan. I downloaded the coordinates for a succinate 3D conformer from PubChem and imported and positioned it for reference (green carbons) [22]. This image was taken looking down over the extracellular side of the receptor. For reference and clarity, extracellular loop 2 (EL2) is colored magenta because it partially covers the binding area in the current conformation. Key transmembrane α-helices were labeled after the fact (for later reference) according to Figure 3(c) by *He et al.* and Figure 1 by *Costanzi* [26,27]:



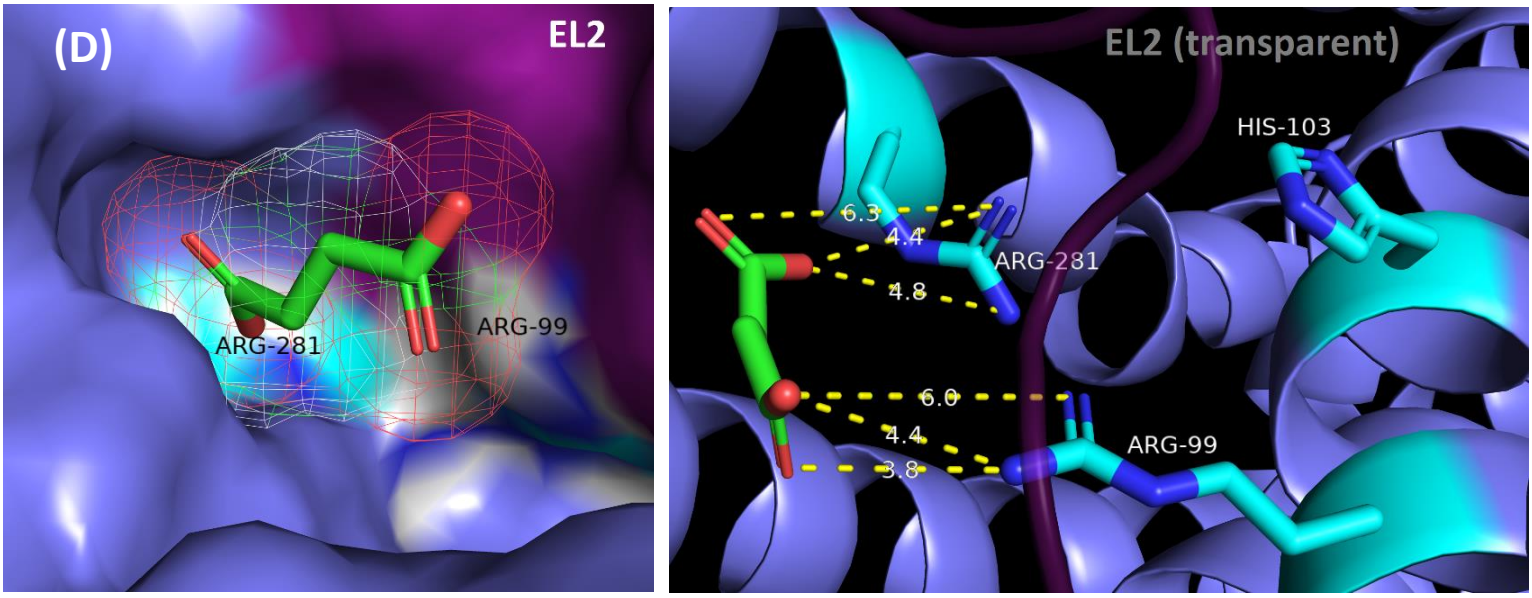
Based on the image shown on the previous page, I could imagine the 4 highlighted residues contributing to the sensing of extracellular succinate by enabling it to bind, which would initiate larger conformational changes in hSUCNR1. Changes on the extracellular side would be triggered by succinate binding and forcing shifts in the relative positions of certain α-helices. Because the helices run the length of the cell membrane, changes on their extracellular side would likely translate to conformational shifts on the *intracellular* side. I could imagine succinate's interactions with these residues "pulling" the extracellular ends of TM6 and/or TM7 closer to TM3 (or vice versa), or even pulling those ends of TM6 and TM7 closer together directly, depending on where and how succinate binds. This could cause some sort of "molecular lever action", in that when the ends of a set of α-helices are pulled together, the opposite ends might spread apart because they are still somewhat anchored in the middle by their interactions with other nearby helices. While I say the ends of certain are α-helices "pulled together" causing a "lever-like" mode of conformation effects on the other end, this is really just an analogy and the specifics could be quite different, such as succinate inducing the *rotation* of certain helices (or other changes). The point is that the affected helices' interactions with each other around the succinate binding site ultimately translate to shifts in position on their cytosolic ends, which enable SUCNR1 to bind/activate G-proteins. Also, while I am focusing on TM3, TM6, and TM7 because they contain the residues in question, I could see shifts in any of these helices contributing to conformational changes in their neighbors (e.g. TM5 being affected by TM6). Whatever the specifics of the structural changes initiated by succinate binding to these residues, the result is that they put the intracellular side of SUCNR1 into an active conformation where its structure can facilitate the activation of its associated G proteins. Below is a crude representation of some of the possible modes of signal transduction through SUCNR1 proposed above:



It is worth noting that **Figure A & B** above only provide one example of conformation and positioning of succinate and the side chains (rotamers), and there are likely degrees of freedom in either the ligand or side chains for a variety of specific charge-based binding scenarios. The point is that whichever of these residues succinate binds to, it will likely be "bridging a gap" or otherwise causing forces to be exerted on the helices involved, leading to further adjustments and conformational changes that impact both sides of the receptor. Another important thing to highlight is that **EL2 was hidden in the images above**. The homology models produced by all 3 methods earlier have EL2 in a conformation that blocks access to ARG-252 and HIS-103. This can be seen in **Figure C** at the top of the next page, which shows the surface representation of the RosettaCM model for hSUCNR1 with and without the EL2 loop included in the surface representation in PyMol.



Clearly, results from the mutation experiment by *He et al.* displayed in the HW3 instructions show that HIS-103 and ARG-252 are necessary for hSUCNR1 activation [26]. Perhaps the EL2 loop often adopts a more open conformation than the one in my models, allowing access to these residues by succinate. Either that or HIS-103 and ARG-252 play a different, indirect role that is still essential for activation. Even with EL2 in the conformation shown above in **Figure C**, ARG-99 and ARG-281 still appear to be accessible by succinate in the RosettaCM model. Given the likely rotational/conformational freedom of succinate and the side chains, I think succinate may be able to bind with those two residues despite this EL2 conformation. The image below shows an approximate possible ionic binding scenario with succinate, ARG-99, and ARG-281 in this less open pocket conformation. **Figure D** was prepared by manually editing succinate's bond rotations and overall rotation/position until the succinate carboxylate groups were as close to the arginine side chains as possible with minimal surface-surface overlap (no side chain rotamers were altered). Even taking this crude approach, it is clear that succinate can likely fit in the pocket enough to initiate electrostatic interactions:



Perhaps even in the "closed" EL2 conformation, an interaction like the one depicted in **Figure D** above involving ARG-99 and ARG-281 could initiate changes in the EL2 loop position, possibly enabling access and further interactions with ARG-252 and/or HIS-103 by succinate. This could contribute to the same general process discussed earlier, causing shifts in TM6 and TM7 (and possibly even TM3 and/or TM5) around the binding site,

which then lead to conformational changes on the intracellular side of the affected helices. Again, such changes could ultimately enable hSUCNR1 to bind the relevant G-proteins and activate them.

Given the fact that structures of this receptor have recently been submitted, I was curious what the associated publications had to say about hSUCNR1 activation by succinate. The group that submitted the Cryo-EM structure of hSUCNR1 discussed briefly at the end of Question 1 (8YKV) investigated the binding and activation of this receptor by several substrates [18]. Interestingly, in their structure, they only observed ionic binding with ARG-99 and ARG-281, along with hydrogen bonds with two nearby tyrosine side chains [18]. This is consistent with my observation above that these were the only two accessible residues out of the four considered, at least in my homology models. It is worth noting they did observe HIS-103 being involved in binding with two different ligands [18]. By comparing their activated hSUCNR1 structures to that of inactivated rat SUCNR1, they observed shifts on the intracellular sides of TM5, TM6, and TM7, as well as in the helix perpendicular to the end of TM7 (see Figure 1(m) by *Li et al.*) [18]. The authors said that, on the intracellular side of the receptor, the outward movement of TM6 and the inward movement of TM7 gives room for the α5 helix of one of the G-protein's subunits (Gαi) to bind this receptor [18]. The authors of another recent paper still identified ARG-281 as a residue that interacts directly with succinate in their Cryo-EM structure, but they did not find direct interactions with ARG-99, HIS-103, or ARG-252 [28]. They also reported that succinate binding leads to outward shifts in TM5 and TM6 on the intracellular side, as well as shifts in TM7 and its attached perpendicular helix (H8) [28]. Again, they state that these shifts make way for an α-helix (α5) of a G-protein subunit (Gαi) to pack against part of the intracellular side of TM6 [28]. Thus, the general notion is that succinate binding induces key changes to the intracellular positioning of a few α-helices to allow G-protein binding [18,28]. What I proposed earlier in **Figures A and B** was much vaguer than the specifics described by *Liu et al.* (see Figure 4 in that paper) [28]. Succinate binding appears to cause repositioning of certain side chains on the affected helices, which leads to the breaking of some interactions and the formation of new ones [28]. More specifically, *Liu et al.* propose that the binding of ligands to hSUCNR1 may pull TM3 inward and break a hydrophobic core, which then forces TM5 and TM6 outward, particularly on their intracellular ends [28]. For me, these papers highlighted how challenging it can be to identify plausible specifics of signal transduction through the structure of a receptor such as hGPR91, even when models of the active and inactive states are available. It was also interesting to see that everyone seems to propose overlapping but slightly different sets of residues that directly interact with succinate [18,26,28]. This emphasizes the sensitivity of our conclusions to the specifics of the structure(s) we have on hand. This is further complicated by the fact that all structures come with uncertainty due to their resolution (if solved) or how they were built (if predicted). Given the study by *Costanzi*, I think the possible flexibility/freedom of that EL2 loop may partially explain the variability in the specific residues researchers identify as being directly involved in interactions with succinate [15,18,26,28]. Mutation experiments can provide strong evidence for which residues are essential for SUCNR1's function, but it may be difficult to tease apart which of those are essential due to direct binding versus indirect roles that are still necessary for activation (e.g. by facilitating structural changes and/or stability in the activated state).

References:

[1] https://en.wikipedia.org/wiki/List_of_protein_structure_prediction_software

[2] Kelley, L. A., Mezulis, S., Yates, C. M., Wass, M. N., & Sternberg, M. J. E. (2015). The Phyre2 web portal for protein modeling, prediction and analysis. In Nature Protocols (Vol. 10, Issue 6, pp. 845–858). Springer Science and Business Media LLC. https://doi.org/10.1038/nprot.2015.053

[3] Song, Y., DiMaio, F., Wang, R. Y.-R., Kim, D., Miles, C., Brunette, T., Thompson, J., & Baker, D. (2013). High-Resolution Comparative Modeling with RosettaCM. In Structure (Vol. 21, Issue 10, pp. 1735–1742). Elsevier BV. https://doi.org/10.1016/j.str.2013.08.005

[4] Zimmermann, L., Stephens, A., Nam, S.-Z., Rau, D., Kübler, J., Lozajic, M., Gabler, F., Söding, J., Lupas, A. N., & Alva, V. (2018). A Completely Reimplemented MPI Bioinformatics Toolkit with a New HHpred Server at its Core. In Journal of Molecular Biology (Vol. 430, Issue 15, pp. 2237–2243). Elsevier BV. https://doi.org/10.1016/j.jmb.2017.12.007

[5] Webb, B., & Sali, A. (2016). Comparative Protein Structure Modeling Using MODELLER. In Current Protocols in Protein Science (Vol. 86, Issue 1). Wiley. https://doi.org/10.1002/cpps.20

[6] https://www.ncbi.nlm.nih.gov/gene/56670#reference-sequences

[7] Haffke, M., & Jaakola, V.-P. (2019). Crystal structure of the rat isoform of the succinate receptor SUCNR1 (GPR91) in complex with a nanobody. Worldwide Protein Data Bank. https://doi.org/10.2210/pdb6ibb/pdb

[8] Krieger, E., Nabuurs, S. B., & Vriend, G. (2003). Chapter 16: HOMOLOGY MODELING. In Structural bioinformatics (Hoboken, NJ: Wiley-Blackwell). pp. 507–519.

[9] Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., & Madden, T. L. (2009). BLAST+: architecture and applications. In BMC Bioinformatics (Vol. 10, Issue 1). Springer Science and Business Media LLC. https://doi.org/10.1186/1471-2105-10-421

[10] Haffke, M., & Jaakola, V.-P. (2019). Crystal structure of a humanized (K18E, K269N) rat succinate receptor SUCNR1 (GPR91) in complex with a nanobody and antagonist NF-56-EJ40. Worldwide Protein Data Bank. https://doi.org/10.2210/pdb6rnk/pdb

[11] Steinegger, M., Meier, M., Mirdita, M., Vöhringer, H., Haunsberger, S. J., & Söding, J. (2019). HH-suite3 for fast remote homology detection and deep protein annotation. In BMC Bioinformatics (Vol. 20, Issue 1). Springer Science and Business Media LLC. https://doi.org/10.1186/s12859-019-3019-7

[12] https://toolkit.tuebingen.mpg.de/tools/modeller (text found by clicking the "?")

[13] https://salilab.org/modeller/

[14] https://en.wikipedia.org/wiki/Homology_modeling#Satisfaction_of_spatial_restraints

[15] Costanzi, S. (2008). On the Applicability of GPCR Homology Models to Computer-Aided Drug Discovery: A Comparison between In Silico and Crystal Structures of the β2-Adrenergic Receptor. In Journal of Medicinal Chemistry (Vol. 51, Issue 10, pp. 2907–2914). American Chemical Society (ACS). https://doi.org/10.1021/jm800044k

[16] Williams, C. J., Headd, J. J., Moriarty, N. W., Prisant, M. G., Videau, L. L., Deis, L. N., Verma, V., Keedy, D. A., Hintze, B. J., Chen, V. B., Jain, S., Lewis, S. M., Arendall, W. B., III, Snoeyink, J., Adams, P. D., Lovell, S. C., Richardson, J. S., & Richardson, D. C. (2017). MolProbity: More and better reference data for improved all-atom structure validation. In Protein Science (Vol. 27, Issue 1, pp. 293–315). Wiley. https://doi.org/10.1002/pro.3330

[17] Li, C., Liu, H., Li, J., Zhu, H., Fu, W., & Xu, H. E. (2024). Cryo-EM structure of succinate receptor SUCR1 bound to compound 31. Worldwide Protein Data Bank. https://doi.org/10.2210/pdb8ykv/pdb

[18] Li, C., Liu, H., Li, J., He, X., Zhu, H., Fu, W., & Xu, H. E. (2024). Molecular basis of ligand recognition and activation of the human succinate receptor SUCR1. In Cell Research (Vol. 34, Issue 8, pp. 594–596). Springer Science and Business Media LLC. https://doi.org/10.1038/s41422-024-00984-7

[19] Haffke, M., Fehlmann, D., Rummel, G., Boivineau, J., Duckely, M., Gommermann, N., Cotesta, S., Sirockin, F., Freuler, F., Littlewood-Evans, A., Kaupmann, K., & Jaakola, V.-P. (2019). Structural basis of species-selective antagonist binding to the succinate receptor. In Nature (Vol. 574, Issue 7779, pp. 581–585). Springer Science and Business Media LLC. https://doi.org/10.1038/s41586-019-1663-8

[20] https://en.wikipedia.org/wiki/Succinic_acid

[21] https://en.wikipedia.org/wiki/Carboxylate

[22] https://pubchem.ncbi.nlm.nih.gov/compound/Succinate

[23] https://en.wikipedia.org/wiki/Amino_acid

[24] https://en.wikipedia.org/wiki/Arginine

[25] https://www.imgt.org/IMGTeducation/Aide-memoire/_UK/aminoacids/charge/#:~:text=Table%201%3A%20pK%20of%20the,or%20amine%20group%20is%20charged.&text=The%20arginine%20side%20chain%20is,charge%20is%20stabilized%20by%20resonance.

[26] He, W., Miao, F. J.-P., Lin, D. C.-H., Schwandner, R. T., Wang, Z., Gao, J., Chen, J.-L., Tian, H., & Ling, L. (2004). Citric acid cycle intermediates as ligands for orphan G-protein-coupled receptors. In Nature (Vol. 429, Issue 6988, pp. 188–193). Springer Science and Business Media LLC. https://doi.org/10.1038/nature02488

[27] Costanzi, S. (2008). On the Applicability of GPCR Homology Models to Computer-Aided Drug Discovery: A Comparison between In Silico and Crystal Structures of the β2-Adrenergic Receptor. In Journal of Medicinal Chemistry (Vol. 51, Issue 10, pp. 2907–2914). American Chemical Society (ACS). https://doi.org/10.1021/jm800044k

[28] Liu, A., Liu, Y., Zhang, W., & Ye, R. D. (2024). Structural insights into ligand recognition and activation of the succinate receptor SUCNR1. In Cell Reports (Vol. 43, Issue 7, p. 114381). Elsevier BV. https://doi.org/10.1016/j.celrep.2024.114381